Ryan M. C. Law

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Education

PhD in Medical Science 2022 – 2026
MRC Cognition and Brain Sciences Unit, University of Cambridge, UK
Supervisors: Dr Olaf Hauk & Prof Matthew A. Lambon Ralph

MSc in Cognitive Neuroscience (Research) 2021 – 2022
Donders Institute for Brain, Cognition and Behaviour, Radboud University, NL
Supervisors: Dr Hugo Weissbart & Dr Andrea E. Martin

BA in Linguistics 2015 – 2018
University College London, UK
Supervisor: Dr Wing Yee Chow

Research positions

Research project coordinator, Max Planck Institute for Psycholinguistics 2020 – 2022

Predoctoral researcher, New York University 2018 – 2020

Awards and Scholarships

Travel Grant, 20th Alpine Brain Imaging Meeting (CHF1,000) 2026

Travel Grant (shortlisted), Society for the Neurobiology of Language 2025

Diverse Intelligences Summer Institute Project Accelerator Grant ($3,000) 2024

Student Award, 23rd International Conference on Biomagnetism (AUD500) 2024

Diverse Intelligences Summer Institute Fellow, St Andrews, UK. 2024

Gates Cambridge Scholarship (Full PhD funding at Overseas rate, ~£250k; *accepted*), Cambridge, UK. 2022

Leverhulme Doctoral Training Programme for the Ecological Study of the Brain (*declined*). 2022

UCL Overseas Research Scholarship (PhD tuition at Overseas rate; *declined*). 2022

UCL Graduate Research Scholarship (PhD tuition at UK Home rate and maintenance; *declined*). 2022

UCL-Birkbeck Medical Research Council Doctoral Training Programme Funding (PhD tuition at UK Home rate and maintenance; *declined*). 2022

UCL International Scholar Award for Doctoral Training (PhD tuition at Overseas rate; *declined*). 2022

Featured Scientist, 100 Scientists of Malaysia, Malaysia. 2021

Best Presentation Prize, UCL Faculty of Brain Sciences Undergraduate Conference. UCL, UK. 2018

Sultan Ibrahim Johor Bursary, State of Johor, Malaysia. 2018
Awarded to promising Malaysian students studying abroad from the Sultanate of Johor in Malaysia.

Summer Undergraduate Research Internship Bursary, Harvard University, USA. 2017

UCL International Short Mobility Bursary, UCL, UK. 2017

Award for Academic Excellence, UCL Department of Linguistics, UK. 2015

Publications

Peer-reviewed articles

[1] **Law, R.M.C.**, Lambon Ralph, M.A., Hauk, O. (2026). A common framework for semantic memory and semantic composition. *Imaging Neuroscience*. [DOI]

[2] Rane, S.*, Kirkman, C.F.*, Todd. G.*, Royka, A.*, **Law, R.M.C.***, Cartmill, E., Foster, J.G. (2025). Position: Principles of Animal Cognition to Improve LLM Evaluations. *Proceedings of the 42nd International Conference on Machine Learning (ICML).* (*equal contribution.) [OpenReview]

[3] Hauk, O., Magnabosco F., **Law, R.M.C.** (2023). Can we separate semantic representations from computations? A commentary on Calzavarini (2023). *Language, Cognition and Neuroscience*. [DOI]

[4] **Law, R.**, Pylkkänen, L. (2021). Lists with and without syntax: A new approach to measuring the neural processing of syntax. *Journal of Neuroscience.* [DOI]

[5] Tulling, M., **Law, R.**, Cournane, A., Pylkkänen, L. (2020). Neural correlates of modal displacement and discourse-updating under (un)certainty. *eNeuro.* [DOI]

Manuscripts under review

[1] **Law, R.M.C.**, Hauk, O., Lambon Ralph, M.A. Anterior temporal lobes show route-dependent sensitivity to conceptual specificity: Evidence from MEG and EEG. [preprint]

Manuscripts in preparation

[1] **Law, R.M.C.**, Rogers, T.T., Lambon Ralph, M.A. (*in prep*). A dual-pathway neural network architecture for item-independent structure learning from semantic and arbitrary sequences.

[2] **Law, R.**, Weissbart, H., Martin, A.E., (*in prep*). Representational dynamics of meaning composition during natural listening.

Presentations

Invited talks

[1] *Norwegian University of Science and Technology*, Language Acquisition and Language Processing Lab. Trondheim, Norway. (2026, September).

[2] *University of Geneva, NeuroLingo Lab*. Geneva, Switzerland. (2026, January)

[3] *University of Oxford, Psycholinguistics Seminar*. Oxford, UK. (2025, May)

[4] *New York University*, *Neuroscience of Language Lab*, USA. (2025, May)

[5] *University of Cambridge, Cambridge Neuroscience*, Cambridge, UK. (2025, March).

[6] *UCL, Linguistics Departmental Seminar Series*. London, UK. (2024, May).

[7] *Bristol University, Department of Computer Science*. Bristol, UK. (2023, December).

[8] *Dartmouth College, Department of Linguistics*. Hanover, New Hampshire, USA. (2020, October).

Slide presentations

[1] **Law, R.M.C.**, Hauk, O., Lambon Ralph, M.A. (2026, January). Single word conceptual specificity, not phrasal composition, drives anterior temporal lobe during semantic processing: evidence from combined EEG/MEG. Blitz talk presentation at the 20th Alpine Brain Imaging Meeting 2026. Champéry, Switzerland.

[2] **Law, R.M.C.**, Lambon Ralph, M.A., Hauk, O. (2025, September). A common framework for semantic memory and semantic composition. Slide presentation at the Society for the Neurobiology of Language 17th Annual Meeting. Washington DC, USA.

[3] **Law, R.M.C.**, Hauk, O., Lambon Ralph, M.A. (2025, July). Meaning construction in the anterior temporal lobe: an MEG/EEG study. Blitz talk presentation at the MEG-UKI Conference 2025. London, UK.

[4] Rane, S.*, Kirkman, C.F.*, Todd. G.*, Royka, A.*, **Law, R.M.C.***, Cartmill, E., Foster, J.G. (2024, December). Principles of Animal Cognition for LLM Evaluations: A Case Study on Transitive Inference. *NeurIPS 2024 Workshop on Behavioral Machine Learning*. Vancouver, Canada. (*equal contribution.)

[5] **Law, R.M.C.**, Lambon Ralph, M.A., Hauk, O. (2024, August). Tracking compositional semantic processing using MEG and EEG. Slide presentation at the 23rd International Conference on Biomagnetism. Sydney, Australia.

[6] **Law, R.**, Weissbart, H., Martin, A.E. (2022, May). Decoding composition during natural listening. Slide presentation at the Radboud University Cognitive Neuroscience Conference Synapsium. Nijmegen, the Netherlands.

[7] **Law, R.**, Kaufeld, G., Bosker, H.R., Martin, A.E. (2021, September). Cortical tracking of linguistic units at different rates. Slide slam at 13th Annual Meeting of the Society for the Neurobiology of Language Virtual Edition.

[8] **Law, R.**, Pylkkänen, L. (2020, May). Lists with and without syntax: Neural correlates of syntactic structure. Data blitz presentation at the CNS 2020 Virtual Meeting.

[9] **Law, R.**, Pylkkänen, L. (2019, April). Neural correlates of syntactic structure: MEG study using lists. Slide presentation at the Neuroscience of Language Conference 2019. Abu Dhabi, UAE.

[10] **Law, R.**, Arkhipova, Y.**,** Chow, W.Y. (2018, March). Tracking expectation reversal during discourse comprehension using EEG. Slide presentation at the Faculty of Brain Sciences Fifth Annual Undergraduate Conference. London, UK.

Poster presentations

[1] **Law, R.M.C.**, Rogers, T.T., Lambon Ralph, M.A. (2026, October). A theory of number and magnitude learning. Poster presentation at the 18th Annual Meeting of the Society for the Neurobiology of Language. Geneva, Switzerland.

[2] **Law, R.M.C.**, Hauk, O., Lambon Ralph, M.A. (2026, January). Single word conceptual specificity, not phrasal composition, drives anterior temporal lobe during semantic processing: evidence from combined EEG/MEG. Poster presentation at the 20th Alpine Brain Imaging Meeting 2026. Champéry, Switzerland.

[3] **Law, R.M.C.**, Hauk, O., Lambon Ralph, M.A. (2025, July). Meaning construction in the anterior temporal lobe: an MEG/EEG study. Poster presentation at the MEG-UKI Conference 2025. London, UK.

[4] **Law, R.M.C.**, Lambon Ralph, M.A., Hauk, O., (2024, October). Tracking the neural dynamics of compositional semantics with MEG and EEG. Poster presentation at the MEG-UKI Conference 2024. Birmingham, UK.

[5] **Law, R.M.C.**, Lambon Ralph, M.A., Hauk, O., (2023, October). An MEEG study bridging semantic composition and semantic cognition. Poster presentation at the 15th Annual Meeting of the Society of the Neurobiology of Language. Marseille, France.

[6] **Law, R.**, Weissbart, H., Martin, A.E., (2022, June). Decoding composition during natural listening. Poster presentation at the International Max Planck Research School Conference: Interdisciplinary Approaches to the Language Sciences. Nijmegen, the Netherlands.

[7] **Law, R.**, Kaufeld, G., Bosker, H.R., Martin, A.E., (2022, April). Cortical tracking of linguistic units at different rates. 18th NVP Winter Conference on Brain and Cognition. Egmond aan Zee, the Netherlands.

[8] **Law, R.**, Pylkkänen, L. (2020, June). Lists with and without syntax: Neural correlates of syntactic structure. Virtual poster presentation at the International Max Planck Research School Conference: Interdisciplinary Approaches to the Language Sciences.

[9] **Law, R.**, Pylkkänen, L. (2020, May). Lists with and without syntax: Neural correlates of syntactic structure. Poster presentation at the CNS 2020 Virtual Meeting.

[10] **Law, R.**, Pylkkänen, L. (2019, October). Lists with and without syntax: MEG effects of syntactic composition when local conceptual combination is controlled. Poster presentation at the International Brain and Syntax Think Tank. Evanston, USA.

[11] Arkhipova, Y., Sostarics, T., **Law, R.**, Xiang, M., Chow, W.Y. (2019, September). 'Nonetheless' can reverse predictions immediately: evidence from ERPs. Poster presentation at the 25th Architectures and Mechanisms of Language Processing Conference. Moscow, Russia.

[12] **Law, R.**, Pylkkänen, L. (2019, August). Lists with and without syntax: MEG effects of syntactic composition when local conceptual combination is controlled. Poster presentation at the 11th Annual Meeting of the Society for the Neurobiology of Language. Helsinki, Finland.

[13] Tulling, M., **Law, R.**, Cournane, A., Pylkkänen, L. (2019, August). Discourse belief updating in the right hemisphere. Poster presentation at the 11th Annual Meeting of the Society for the Neurobiology of Language. Helsinki, Finland.

[14] Arkhipova, Y., **Law, R.**, Xiang, M., Chow, W.Y. (2019, March). 'Nonetheless' can reverse predictions immediately: evidence from ERPs. Poster presentation at the 32nd Annual CUNY Human Sentence Processing Conference. Boulder, USA.

Public engagement & outreach

- "*Neon-striped cats and other complex concepts in the human brain*". CamBrain Neurotalks, Cambridge (2026, March).
- "*How does the brain understand language?*" Norwich Science Festival (2025, February).

Teaching, supervision & mentorship

Supervisor, Psychology of Language Processing & Learning, University of Cambridge — 2024 – present

Mentor, Cambridge Access to Mental Health & Psychology, Department of Psychiatry, University of Cambridge (aimed at graduate school applicants from underrepresented backgrounds in Psychology and Psychiatry) — 2022

Guest lecturer, Psycholinguistics, Dartmouth College, USA — 2020

Instructor, MEG preprocessing and analysis workshop, New York University Abu — 2020

Dhabi, UAE

Professional and Academic Service

Ad-hoc reviewer for Language, Cognition, and Neuroscience, Topics in Cognitive Science, Journal of Psycholinguistic Research, Cognitive Science Conference

Program Committee member, 18th Annual Meeting of the Society for the Neurobiology of Language — 2026
Chaired a symposium; Lead organiser of 3 consecutive networking events "Lunch with a Professor" and a social event for early-career researcher, securing a USD2,000 budget.

Elected student and post-doc representative to the Board, Society for the Neurobiology of Language — 2025 – present
Advocate for early career researchers to the Board.

Equity, Diversity, and Inclusion Committee member, MRC Cognition and Brain Science Unit, University of Cambridge — 2023 – present

Diversity, Equity, and Inclusion Committee member, Max Planck Institute for Psycholinguistics — 2020–2022

Conference organising committee, Neuroscience of Language Conference Abu Dhabi — 2019

Skills

Research methods: MEG & EEG (acquisition, source reconstruction, multivariate decoding), eye-tracking, behavioural experimental design, neural network modelling

Neuroimaging & open-science tools: MNE-Python, HPC/Slurm pipelines, GitHub, reproducible-research standards (OSF, The Turing Way)

Programming languages: Python (8 years), R, MATLAB, JavaScript

Natural languages: English • Mandarin • Malay • Cantonese • Teochew • Hokkien • Dutch (B1) • Arabic (A2)